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36 (SEM-II) BT 121

2026

MODERN BIOLOGY

Paper : BT-121

Full Marks : 100

Time : Three hours

The figures in the margin indicate full marks for the questions.

1. Select the correct answer : [L1, L2]
1×10=10

(i) RNA differs from DNA in having :

- (A) Thymine
- (B) Deoxyribose sugar
- (C) Uracil
- (D) Double helix structure

(ii) The site of protein synthesis is :

- (A) Nucleus
- (B) Ribosome

- (C) Lysosome
(D) Golgi apparatus
- (iii) The genetic code is :
(A) Overlapping
(B) Non-specific
(C) Degenerate
(D) Ambiguous
- (iv) Which RNA carries amino acids to ribosomes ?
(A) mRNA
(B) tRNA
(C) rRNA
(D) snRNA
- (v) In gel electrophoresis, DNA moves toward :
(A) Cathode
(B) Anode
(C) Neutral side
(D) Does not move

(vi) Which of the following is not a stop codon?

(A) UAA

(B) UAG

(C) UGA

(D) AUG

(vii) Which of the following is a saprophyte?

(A) Bacteria

(B) Fungi

(C) Algae

(D) Protozoa

(viii) BLAST is a tool used for:

(A) Designing primers

(B) Editing genes

(C) Diagnosing diseases

~~(D)~~ Creating databases

(ix) The size of bacteria is in the range of:

(A) $10^{-6}m$

~~(B)~~ $10^{-9}m$

(C) $10^{-3}m$

(D) $10m$

(x) The number of peptide bonds in a tripeptide is :

(A) 2

(B) 3

(C) 1

(D) 4

2. Fill in the blanks : ✓

[L1, L2]
 $1 \times 20 = 20$

(i) Organisms composed of cells that contain nuclei are classified as eukaryoti whereas organisms composed of cells that lack nuclei are classified as prokaryo

(ii) The technique involved in the discovery of the structure of DNA was X Ray.

(iii) ligase is the enzyme that helps in joining the DNA strands.

(iv) The DNA strands are held together by H-bonding

(v) The first mRNA codon in a polypeptide chain is _____.

(vi) A region of the chromosome that harbours information necessary to specify the sequence of amino acids in a particular protein is called a _____.

(vii) Cin

(viii) Ge
spe
an

(ix) On
me
coc

(x) Th
olig

(xi) The
isola

(xii) The
app

(xiii) The
regu

(xiv) The
Li

(xv) PO
infor

(xvi) Micr

(xvii) gl

- (vii) Circular DNA is found in Bacterium.
- (viii) Genetic code is unambiguous i.e. one codon specifies only one amino acid and not any other.
- (ix) Only two amino acids—tryptophan and methionine are specified by single codon, UAC for tryptophan and AUG for methionine.
- (x) The number of sugar units in oligosaccharides range from 2 to 10 to _____.
- (xi) The enzyme reverse transcriptase was isolated from retro.
- (xii) The human genome contains approximately 3 billion base pairs.
- (xiii) The hormone, insulin helps in regulating blood sugar.
- (xiv) The structure of cholesterol has 4 rings.
- (xv) Protein Data Bank (PDB) is a database that stores information on structure of proteins.
- (xvi) Microscope was invented by Nicolaus J. van Leeuwenhoek.
- (xvii) glucose is a reducing sugar.

3✓ Write the full form of the following abbreviations : [L1]
 $2 \times 5 = 10$

- (i) BLAST
- (ii) NCBI
- (iii) NIH
- (iv) EMBL
- (v) ATP

4. Differentiate between : [L2]
 $3 \times 6 = 18$

- (i) Fibrous and Globular protein
- (ii) Catabolic reactions and Anabolic reactions
- (iii) Eukaryotes and Prokaryotes
- (iv) Saturated fats and Unsaturated fats
- (v) Sterilization and Disinfection
- (vi) Primary and Secondary databases

5✓ Answer the following questions : $3 \times 6 = 18$

- (i) Why are lipids insoluble in water but soluble in organic solvents? Explain.
- (ii) Is RNA more stable than DNA?
- (iii) Why does denaturation of proteins lead to loss of function?

(iv)

(v)

(vi)

6. Draw

7. State exam system energy

8. (a) V

(b) Y

9. What structure and c

(iv) Why do saturated fats have higher melting points?

(v) What is an Autoclave used for?

(vi) What are triglycerides?

6. Draw the chemical structure of ATP. [L2]
✓ 4

7. State the *two* laws of Bioenergetics. Give an example energy transfer in a biological system highlighting the different forms of energy involved. ✓ [L2]
5

8. (a) What are sequence databases? [L1]

(b) You need to do a genetic study on the hormone insulin. Name *two* databases each for finding the sequence of the gene that code for insulin and also the sequence of insulin. [L3] ✓
5

9. What are polysaccharides? Compare the structure of the *two* forms of starch, glycogen and cellulose. [L1, L4]
2+8=10